DNT-Scout

Structure-based triage for developmental neurotoxicity screening, with calibrated abstention

2026 Global AI4Science Hackathon · AI + Biology / AI Agent · Team NeuroChip Twin (solo) · Francisco Angulo de Lafuente
Code and data: **github.com/Agnuxo1/dnt-scout** (public, MIT) · SciencePal project:
sciencepal.ai/share/248df9d7-b3aa-4607-8e4c-c4de1a9abc5e

Which untested chemicals deserve a microelectrode-array run — and when is the structural evidence too thin to say?

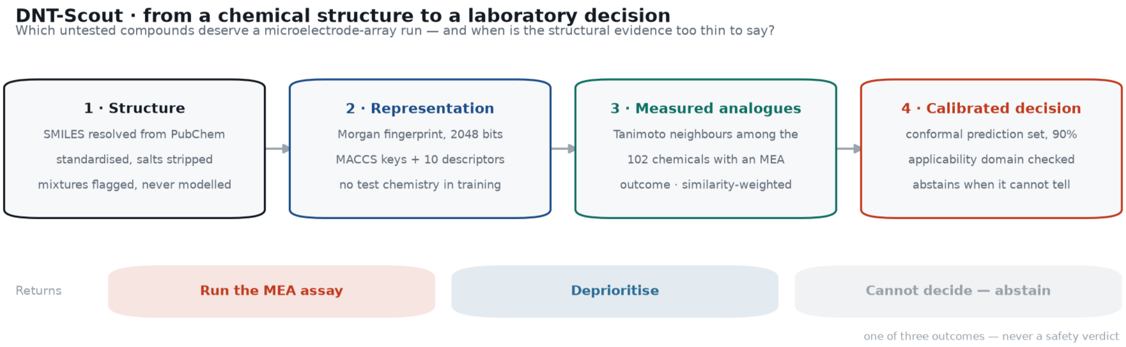
From a chemical structure alone, DNT-Scout ranks compounds by how likely they are to selectively disrupt the formation of neuronal networks, attaches a calibrated statement of confidence, and refuses to answer when the chemistry lies outside what it has evidence for. **32 microelectrode-array runs instead of 51.5** to find half the active compounds — about a third of the laboratory work removed, and 33–34 runs typically across 200 independent splits. An independent audit puts the signal at **p = 0.001** on a 999-permutation test. It is a triage tool for a laboratory queue, not a safety verdict.

1 · The problem, and why it is a scheduling problem

Developmental neurotoxicity is among the most expensive endpoints in regulatory toxicology and one of the least covered: tens of thousands of chemicals are in commerce, and a few hundred have been through a full battery. The in vitro alternative adopted by the OECD DNT-IVB guidance (2023) — growing cortical neurons on a **microelectrode array** and watching whether they organise into a firing network — is far faster than an animal study, but still costs days of culture per compound. With the US EPA moving to phase out mammalian testing, the binding constraint stops being is this chemical neurotoxic? and becomes: **of the compounds nobody has tested, which ones go on the plate first?**

That phrasing matters, because it makes the question measurable. The answer is not an accuracy figure; it is **how many assays you have to run to find the actives**. Every claim here is validated against that number.

The gap. ToxPrint features have been associated with MEA activity without a validated model (Strickland 2018; Kosnik 2020), and conformal prediction has been applied to Tox21 but to no neuronal endpoint (Zhang 2021; Morger 2021). We found no published work doing scaffold-held-out prediction of selective network-formation activity together with calibrated uncertainty and an explicit applicability domain.


DNT-Scout · from a chemical structure to a laboratory decision
Which untested compounds deserve a microelectrode-array run — and when is the structural evidence too thin to say?
1 · Structure
SMILES resolved from PubChem
standardised, salts stripped
mixtures flagged, never modelled
2 · Representation
Morgan fingerprint, 2048 bits
MACCS keys + 10 descriptors
no test chemistry in training
3 · Measured analogues
Tanimoto neighbours among the
102 chemicals with an MEA
outcome · similarity-weighted
4 · Calibrated decision
conformal prediction set, 90%
applicability domain checked
abstains when it cannot tell
Returns
Run the MEA assay
Deprioritise
Cannot decide — abstain
one of three outcomes — never a safety verdict

Four steps. Structures are resolved and standardised; mixtures and inorganics are flagged and excluded rather than coerced. The decision returned is one of three, and never a safety verdict.

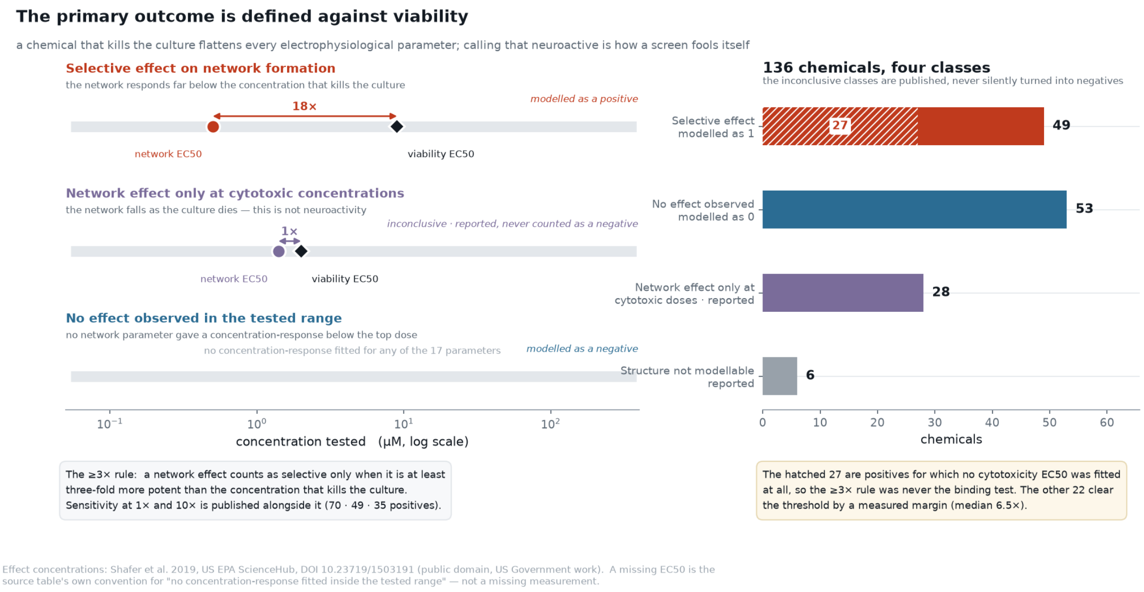
2 · What counts as a hit — the trap this endpoint avoids

A chemical that kills the culture flattens every electrophysiological parameter. Counting that as neuroactive is the main way a developmental-neurotoxicity screen fools itself, so the primary outcome is defined **against viability**: at least one of the 17 network parameters must respond at a concentration at least three-fold below any cytotoxic concentration. Ambiguous chemicals keep their own classes and are **never silently turned into negatives**.

Class	n	Role in the analysis
Selective effect on network formation	49	positive
No effect observed in the tested range	53	negative
Network effect only at cytotoxic concentrations	28	reported, excluded, analysed separately (§4)
Structure not modellable (mixture, inorganic, unresolved)	6	reported, excluded

Credit where it is due. The selectivity ratio is not ours: Shafer et al. publish a per-compound "Selectivity Score" which is exactly this quantity. Recomputing it here from the raw EC50 tables reproduces their published values (Pearson r = 0.9995 on log10 across the 39 compounds that match by name; o,p'-DDT at 244.398 in both), which we use as a check that the data pipeline is right rather than as a result. **What is ours** is turning that ratio into a binary outcome, keeping the chemicals it cannot classify visible, and building a triage decision on top.

What "no cytotoxicity" really means. 27 of the 49 positives have no fitted cytotoxicity EC50 at all, so for them the 3× rule was never the binding test. The top tested concentration is 20 µM throughout, which puts a lower bound on each ratio: **18 of the 27 clear 3× on that bound**, so their selectivity holds inside the tested range whatever the missing value would have been, and **9 are genuinely indeterminate** — §5 re-runs the whole evaluation without them. The other 22 clear the threshold by a directly measured margin (median 6.5×), and the model finds both kinds about equally well (ROC 0.854 and 0.806), so this is one endpoint, not two in disguise. The 3× cut is itself a choice, so the class counts at 1×, 3× and 10× are published together: 70 · 49 · 35.



The primary outcome is defined against viability

a chemical that kills the culture flattens every electrophysiological parameter; calling that neuroactive is how a screen fools itself

The ≥3× rule: a network effect counts as selective only when it is at least three-fold more potent than the concentration that kills the culture. Sensitivity at 1× and 10× is published alongside it (70 · 49 · 35 positives).

The hatched 27 are positives for which no cytotoxicity EC50 was fitted at all, so the ≥3× rule was never the binding test. The other 22 clear the threshold by a measured margin (median 6.5×).

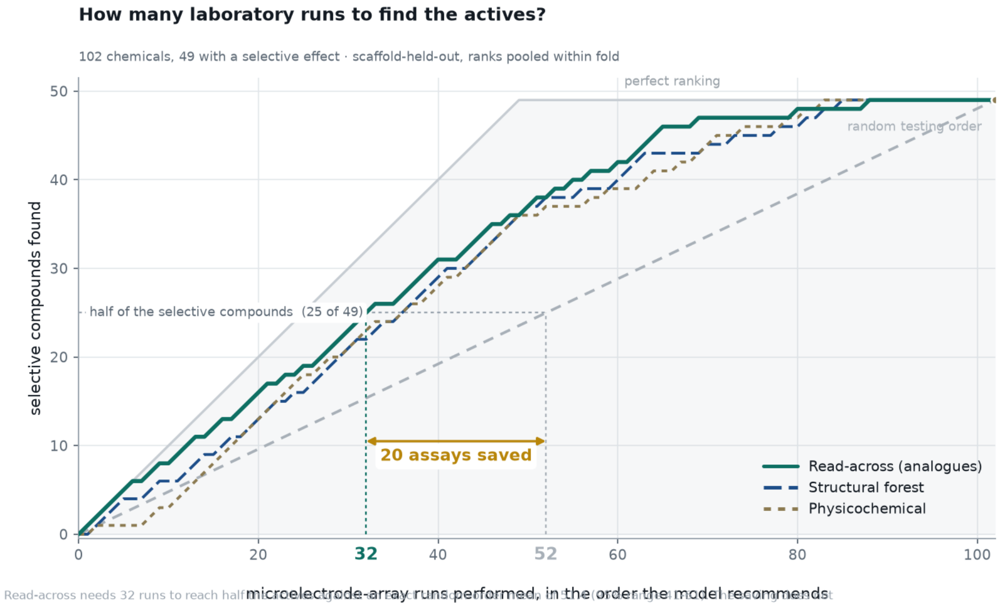
Effect concentrations: Shafer et al. 2019, US EPA ScienceHub, DOI 10.23719/1503191 (public domain, US Government work). A missing EC50 is the source table's own convention for "no concentration-response fitted inside the tested range" — not a missing measurement.

Left: the three archetypes on a concentration axis. Right: the four classes as measured, with the hatched 27 marking the positives the ≥3× rule never actually bound. Source: Shafer et al. 2019, US EPA ScienceHub, DOI 10.23719/1503191 (public domain). A missing EC50 is the source table's own convention for "no concentration-response fitted inside the tested range", not a missing measurement.

3 · Method

Data: 136 chemicals with per-parameter EC50 values across 17 network parameters and two viability assays, re-downloaded from the EPA source and pinned by SHA256; structures from PubChem (132/136), standardised with RDKit. **Models:** a prevalence control; L2 logistic regression on ten physicochemical descriptors; a similarity-weighted k-nearest-analogue **read-across** of the kind US EPA GenRA is built on (re-implemented here, not the EPA service); and a random forest over Morgan fingerprints, MACCS keys and descriptors. Hyperparameters and the conformal calibration split are chosen by grouped cross-validation **inside each training fold**; the test fold is touched once, to predict. Intervals resample whole scaffold groups, never individual chemicals. **Uncertainty:** Mondrian conformal prediction at a nominal 90% with an applicability-domain check, the quantiles and the test scores coming from the same fitted model — which is what makes the guarantee follow.

4 · Result



How many laboratory runs to find the actives?

102 chemicals, 49 with a selective effect · scaffold-held-out, ranks pooled within fold

Read-across needs 32 runs to reach half the actives against an exact random-order mean of 51.5 (95% range 41-62). The saving does not depend on one lucky split: 20 fold assignments give a median of 32 runs (31-37), and an independent audit over 200 fresh assignments finds 33-34 runs typically (29-44).

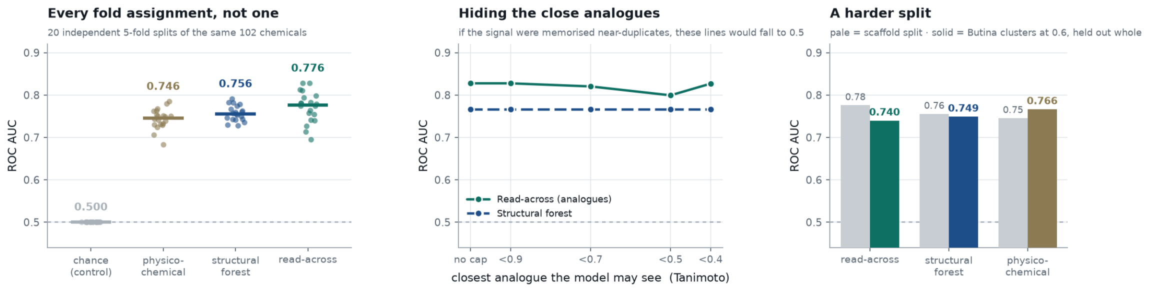
Model	ROC AUC pre-declared split	ROC AUC median of 20 splits	PR AUC	Assays for half the actives
Chance (control)	**0.500**	0.500	0.480	52
Physicochemical descriptors	0.750	0.746	0.635	36
Read-across from measured analogues	**0.828**	**0.776 (0.703–0.828)**	**0.780**	**32 · 33–34 typical**
Structural random forest	0.766	0.756	0.674	36

Half the selective compounds are found after **32** MEA runs on the pre-declared split, against an exact random-order mean of **51.5** (95% range 41–62); across 200 independent splits the figure is 33–34 runs. A perfect oracle would need 25, so the ranking captures roughly three quarters of the gain available at all.

5 · Validation, and four attempts to break the result

An independent audit, and the test that matters most

A separate agent, not sharing our code, reproduced the evaluation from the frozen commit with its own protocol and pre-fixed seeds: **all 185 numeric fields match**, a from-scratch re-implementation agrees to 5.55×10^{-17}, and labels rebuilt independently from the EPA Excel are **136/136 identical**. Its decisive test: **999 label permutations between equal-size scaffolds, with the outer folds, the inner selection and the predictions all refitted each time**. None reached the observed AUC — **$p = 0.001$** (0.003 for the assay budget, 0.004 Bonferroni-corrected). The null it rejects is that structure carries no information about this endpoint. It also found real defects, fixed here, and two that remain open (§9).



Every fold assignment, not one

20 independent 5-fold splits of the same 102 chemicals

Hiding the close analogues

if the signal were memorised near-duplicates, these lines would fall to 0.5

A harder split

pale = scaffold split · solid = Butina clusters at 0.6, held out whole

Left: the spread a single 5-fold split cannot show; the median is the number to quote. Middle and right: two ways of asking whether the model generalises to unfamiliar chemistry rather than recognising close relatives. The structural forest ignores the similarity matrix, so it is flat in the middle panel by construction — an internal control. Ranks are pooled within fold, so a constant predictor scores exactly 0.500.

One split is not evidence. Twenty fold assignments give a median ROC of **0.776 (0.703–0.828)**; the pre-declared split landed at the optimistic end. The audit went further with 200 fresh assignments: median **0.760** under the same grouping rule, and **0.744** when groups are formed without looking at the labels at all — the most conservative figure this result has. The practical number moves less: **33–34 assays typically** (29–44 across 200 splits; 32 on the pre-declared split) against an exact random-order mean of **51.5**, 95% range 41–62.

Is it memorising close relatives? Read-across votes on measured analogues, so this is the obvious worry. Hiding every training analogue above a similarity cap answers it: at a cap of 0.7, ROC 0.820; at 0.5, ROC 0.799. There was little to memorise anyway — the median nearest-analogue Tanimoto is 0.36 and only 4% of chemicals have an analogue above 0.9. The structural forest ignores the similarity matrix and is flat across the cap: the internal control this test needs.

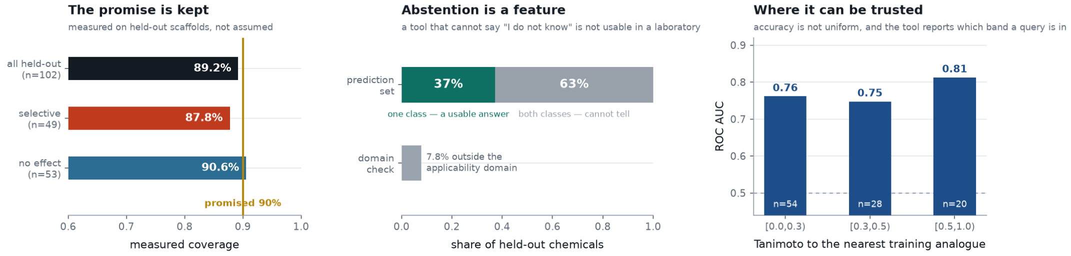
Pooling across folds is a declared choice, not a repair. Five folds produce five models whose probabilities are not on a common scale, so a pooled ranking mixes them: the prevalence control scores 0.473 pooled raw and exactly 0.500 once each score is ranked within its own fold. We report the ranked version and publish all three variants, which agree — read-across is **0.828** ranked, **0.822** as the mean of per-fold AUCs, **0.803** pooled raw. The result does not depend on the choice.

A harder split. 73 of the 81 Bemis-Murcko scaffold groups hold a single chemical — 90% of the groups, 72% of the chemicals — so scaffold-held-out is close to leave-one-out. Holding out whole Butina clusters at 0.6 similarity drops performance to 0.74–0.77 — still far above chance, but the three model families converge. **Read-across being the best model is specific to the scaffold split; structure carrying real signal is not.**

The strictest possible endpoint. The sharpest objection to this work is the 9 positives whose selectivity ratio cannot be bounded above 3×. The whole evaluation was re-run without them, and the result **improves**: read-across reaches ROC **0.832 (0.790–0.877)** and the structural forest **0.839 (0.823–0.871)** on the remaining 93 chemicals, needing **25 assays against 45.8** at random — a 45% saving rather than 38%. Those nine were diluting the signal, not creating it: the headline does not depend on the positives that are hardest to defend, and is stronger without them.

6 · Saying "I don't know"

A triage tool that answers every question is not calibrated, it is guessing. Coverage is **measured on held-out scaffolds, not assumed** — and it belongs to a specific model, which matters because the two here differ. Structural forest: **89.2%** against the 90% promised (87.8% and 90.6% by class). Portable read-across, the one that ships, measured externally by the audit: **94.1%**, median 95.1% over 20 fresh splits — it over-covers, which is the conservative direction, at the cost of more both-class sets. The honest price: the set is uninformative for 63% of chemicals and 7.8% fall outside the applicability domain. Those are the queries where the right answer is a shrug, and the tool gives one.


The promise is kept
measured on held-out scaffolds, not assumed

Abstention is a feature
a tool that cannot say "I do not know" is not usable in a laboratory

Where it can be trusted
accuracy is not uniform, and the tool reports which band a query is in

Mondrian (class-conditional) conformal prediction at a nominal 90% confidence, calibrated inside each training fold on the same fitted model that scores the held-out chemicals, and measured on held-out scaffolds. With 102 held-out chemicals the standard error on a coverage of 0.90 is about 3 points, so the measured value agrees with the promise.

7 · Scope, and what lies outside it

The 28 chemicals whose network effect appears only at lethal concentrations are excluded from the primary comparison, and they are not a random 28: they carry 22 scaffolds that then appear nowhere in the modelled set, and they form a coherent mechanistic family — rotenone, valinomycin, tributyltin, five strobilurins, two statins, chlordane, endosulfan, MPP^+. So the harder question is asked too: **selective activity against everything else, cytotoxicants included.** Read-across scores **0.696 (0.595–0.784)** there, 44 assays against 65.5 at random — lower, as it should be, with the interval still excluding chance. The model does rank several of those cytotoxicants highly, and that is worth reading carefully: permethrin sits at a selectivity ratio of 1.9×, rotenone at 2.1×, just under the three-fold line. A high score is the model correctly recognising real neuroactive chemistry; what structure alone does not resolve is the fine selectivity margin. In the EPA protocol viability is read from the same plate, so such an assay still returns a usable answer.

8 · SciencePal in the core step

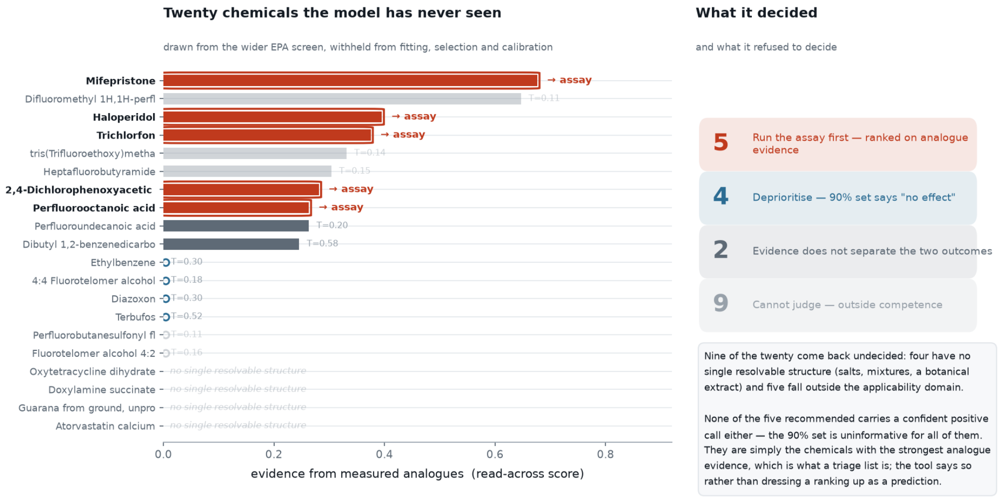
Live thread: **https://sciencepal.ai/share/248df9d7-b3aa-4607-8e4c-c4de1a9abc5e**

The central decision of this project does not happen in our repository — it happens inside SciencePal. The platform's **Biology Agent** is given the 102 measured analogues with their outcomes, 20 chemicals from the wider EPA screen withheld from every stage of fitting, selection and calibration, the project's scoring script and a task specification. Inside the SciencePal sandbox it executes that code, resolves each candidate structure, applies the model and the conformal rule, consults the literature on the nearest analogues, and returns a structured triage report: five chemicals to assay, with evidence, uncertainty, reason and the next assay to run. **Pre-declared ablation:** automatic ranking alone versus ranking + agent on the same 20 withheld chemicals, so the platform's contribution is measured rather than asserted.

	ranking alone	ranking + SciencePal agent
Candidates resolved to a structure	16 / 20	**19 / 20**

With a decisive 90% conformal set	0	**1**
Refusals	9, one bucket	**6, in three distinct categories**
Literature support	none	**4 DOI-cited mechanistic claims**

Result. PubChem was IP-blocked from the SciencePal sandbox. The agent said so instead of failing quietly, routed around it through NCI CACTUS and the US EPA Substance Registry Service, and resolved 19 of 20 against our 16. One of the three it recovered was **doxylamine** — a salt our automated pipeline had discarded, and the only chemical of the twenty with a decisive conformal set (score 1.000). It also **overruled the ranking**, dropping oxytetracycline because its nearest measured analogue is tetracycline at T = 0.50, the highest similarity anywhere in the candidate set, and tetracycline shows no effect — so the score was propped up by two far weaker selective neighbours. Its three headline numbers were recomputed here before being believed, and they reproduce exactly. What the ablation does not establish: the two arms did not face identical network conditions, no accuracy was measured because none of the twenty has a known outcome, and this is one run, not a benchmark.



Nine of the twenty come back undecided: four have no single resolvable structure (salts, mixtures, a botanical extract) and five fall outside the applicability domain.

None of the five recommended carries a confident positive call either — the 90% set is uninformative for all of them. They are simply the chemicals with the strongest analogue evidence, which is what a triage list is; the tool says so rather than dressing a ranking up as a prediction.

The five recommended are the highest-evidence chemicals that pass the applicability-domain check; the ranking, the three nearest measured analogues behind each score and the 90% conformal set are written out for every candidate in results/agent_package/scores.csv, so the recommendation can be audited rather than trusted.

The ranking's own view of the same twenty. Nine come back undecided and none of the five recommended carries a confident positive call — they are the chemicals with the strongest analogue evidence, which is what a triage list is.

9 · Limitations

102 chemicals. Every interval is wide; the per-fold standard deviation of ROC is 0.108. This is a small-data problem and is reported as one. **Cohort prevalence is 48%** because the EPA enriched it with suspected actives, so the assay saving does not transfer unchanged to an unselected library, and we do not claim it does. **Concordance, not external validation:** all 136 labelled chemicals sit inside the wider EPA screen of 460, so the second dataset is a cohort concordance check. **Two inconsistencies found by the audit remain open:** the applicability-domain threshold uses one similarity reference to compute and another to apply (94/102 in domain, 88/102 if made consistent), and the core library and the exported scorer disagree on whether a salt is modellable — neither changes the ranking, both change who gets an answer. **One assay, one laboratory, one developmental window.** **Not a safety verdict:** a negative in vitro result does not exclude developmental neurotoxicity — the OECD guidance says so directly.

10 · Next steps

(1) Prospective validation: the five recommendations in §8 are a falsifiable prediction, and one MEA plate would settle them. **(2)** Extend to the 324 chemicals of the wider screen with network data but no viability measurement — a weaker label, and a natural test of whether the ranking survives a noisier outcome. **(3)** Cross-laboratory concordance between the NTP and ToxCast cohorts inside the EPA screen, and a pre-declared graph-neural-network comparison once there is enough data to make one meaningful.

Originality and resources

All code, figures, results and text were created during the event. The author has a separate, unrelated project (NeuroChip Twin) in another competition; **no code, model, preprocessing, text, figure or video from it is used here** — DNT-Scout is a clean-room project with a new question, repository and code. Data: US EPA network-formation assay (Shafer et al. 2019, DOI 10.23719/1503191) and PubChem, both public domain, re-downloaded and pinned by SHA256; raw data are not redistributed. Software: RDKit, scikit-learn, NumPy, pandas, Matplotlib. **No pretrained models, no proprietary models, no paid APIs.** Claude (Anthropic) assisted with programming and writing under the author's direction; every claim was checked against executed code, and where a result turned out weaker than first reported, the weaker number is the one published. Solo entry; full declaration in **docs/resource_declaration.md**.

Reproducibility: **pip install -e .** then four scripts regenerate every number and figure above; **pytest** runs 23 tests, six of them written to stop the defects in §5 coming back.